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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	CO3

Case Study Topic: : AgroChain – Smart Agriculture Supply Chain Management System.

Work Done Summary:

In this case study, we implemented BFS (Breadth First Search) and DFS (Depth First Search) graph traversal algorithms in the AgroChain Smart Agriculture Supply Chain Management System. The agricultural supply network was represented as a graph where farmers, warehouses, and markets were considered as nodes, and transportation routes were represented as edges.

The BFS algorithm was used to explore all connected locations level by level, while the DFS algorithm was used to analyze complete transportation paths through the network. These algorithms helped in identifying connectivity between different supply chain locations and improving route planning

Implementation Details:

- Represented the agricultural supply chain as a graph.
- Considered farmers, warehouses, and markets as graph nodes.
- Added transportation routes as graph edges.
- Implemented BFS traversal using a queue.
- Implemented DFS traversal using recursion.
- Visited all connected locations in the network.
- Displayed traversal order for supply chain analysis.
- Verified connectivity among all nodes.
- Compared BFS and DFS traversal results.
- Used graph traversal to improve logistics planning.

Input Screenshot :

```
import java.util.*;

public class AgroChainGraph {

    private int vertices;
    private LinkedList<Integer>[] adjList;

    AgroChainGraph(int v) {
        vertices = v;
        adjList = new LinkedList[v];

        for (int i = 0; i < v; i++) {
            adjList[i] = new LinkedList<>();
        }
    }

    void addEdge(int v, int w) {
        adjList[v].add(w);
    }

    void BFS(int start) {

        boolean visited[] = new boolean[vertices];

        Queue<Integer> queue = new LinkedList<>();

        visited[start] = true;
        queue.add(start);
```

```
        if (!visited[n]) {
            visited[n] = true;
            queue.add(n);
        }
    }
}

void DFS(int node, boolean visited[]) {

    visited[node] = true;
    System.out.print(node + " ");

    for (int n : adjList[node]) {
        if (!visited[n]) {
            DFS(n, visited);
        }
    }
}

public static void main(String[] args) {

    AgroChainGraph graph = new AgroChainGraph(5);

    graph.addEdge(0, 1); // Farmer -> Warehouse
    graph.addEdge(1, 2); // Warehouse -> Market A
    graph.addEdge(1, 3); // Warehouse -> Market B
    graph.addEdge(2, 4); // Market A -> Customer

    System.out.println("AGROCHAIN SUPPLY NETWORK");

    System.out.print("\nBFS Traversal: ");
    graph.BFS(0);

    System.out.print("\nDFS Traversal: ");
    boolean visited[] = new boolean[5];
    graph.DFS(0, visited);
}
}
```

Output Screenshot :

```
AGROCHAIN SUPPLY NETWORK

Node Representation:
0 = Farmer
1 = Warehouse
2 = Market A
3 = Market B
4 = Customer

BFS Traversal:
0 1 2 3 4

DFS Traversal:
0 1 2 4 3

Network Connectivity Verified
Process Completed Successfully
```

Result :

The BFS algorithm visited locations level by level and efficiently identified nearby connected locations. The DFS algorithm explored complete transportation paths before moving to the next branch. Both algorithms successfully traversed the agricultural supply network and confirmed connectivity among all supply chain participants.

Time Complexity:

- BFS Traversal: $O(V + E)$
- DFS Traversal: $O(V + E)$

GITHUB LINK : <https://github.com/Himavarshini-07/AgroChain-Smart-Agriculture-Supply-Chain-Management-System>

Student Signature	Faculty Signature